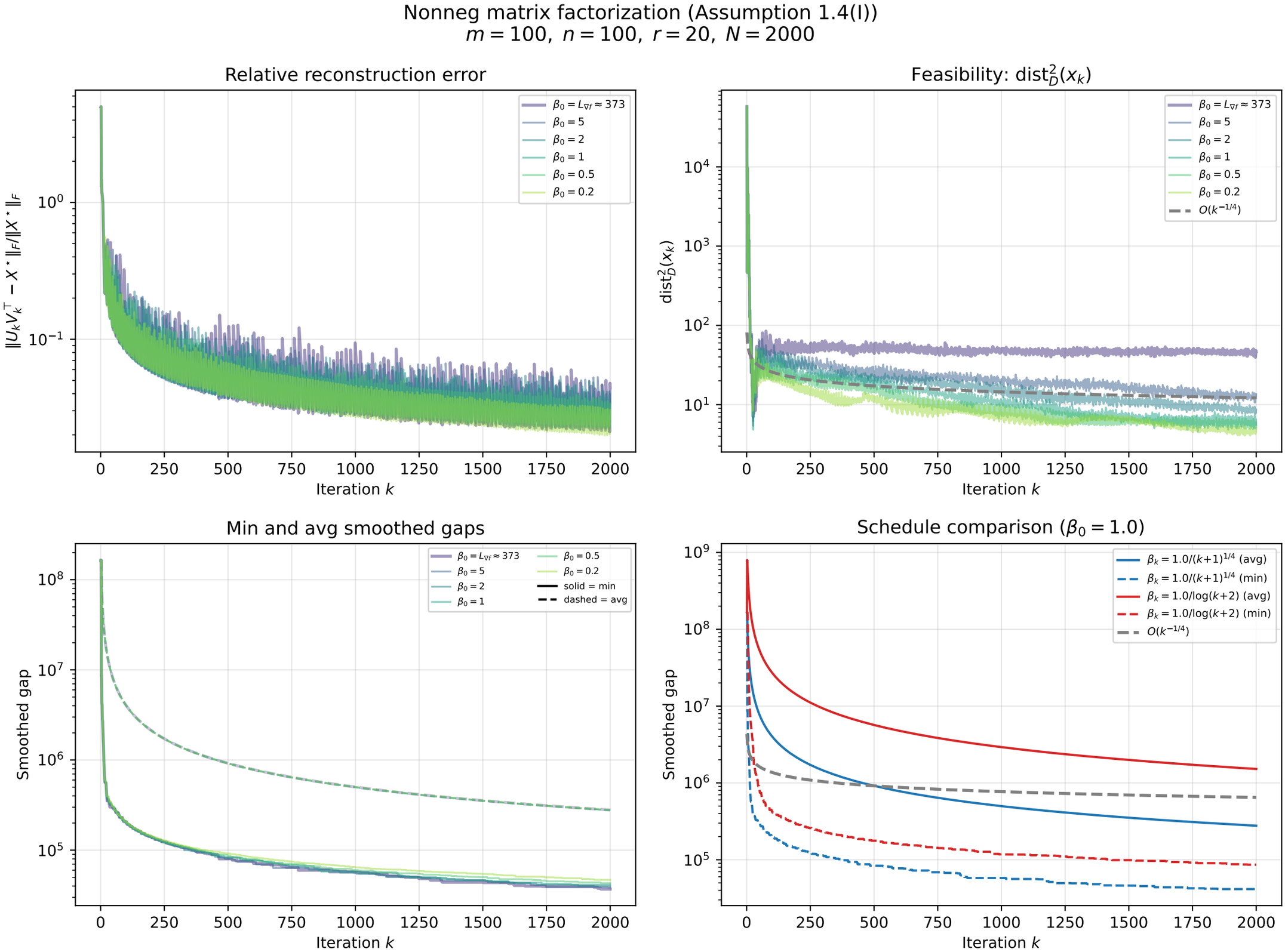
Nonneg matrix factorization (Assumption 1.4(I))
$m = 100,\ n = 100,\ r = 20,\ N = 2000$
Relative reconstruction error
$\|U_k V_k^\top - X^\star\|_F / \|X^\star\|_F$
Iteration k
$\beta_0 = L_{\nabla f} \approx 373$
$\beta_0 = 5$
$\beta_0 = 2$
$\beta_0 = 1$
$\beta_0 = 0.5$
$\beta_0 = 0.2$
Feasibility: $\mathrm{dist}_D^2(x_k)$
$\mathrm{dist}_D^2(x_k)$
$O(k^{-1/4})$
Min and avg smoothed gaps
Smoothed gap
solid = min
dashed = avg
Schedule comparison ($\beta_0 = 1.0$)
$\beta_k = 1.0/(k+1)^{1/4}$ (avg)
$\beta_k = 1.0/(k+1)^{1/4}$ (min)
$\beta_k = 1.0/\log(k+2)$ (avg)
$\beta_k = 1.0/\log(k+2)$ (min)